SECTIONS OF A COLIMIT OF SHEAVES INDEXED BY AN ORDINAL OF LARGE COFINALITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Colimits of sheaves as sheafified colimits of presheaves. Same chapter, Lemma lemma-colimit-sheaves. Identifies the colimit of the system of sheaves with the sheafification of the colimit computed in presheaves.
- The plus construction and sheafification. Same chapter, the description of the sheafification as the result of applying the plus construction twice. Supplies the criterion for a presheaf to be a sheaf used at the end of the proof.

Source reference. Original section 00VB: sources/stacks-sites/pinned-original-source.tex, source lines 3938–3938. Navigation pointer invoked at proof line 3938; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WI: sources/stacks-sites/pinned-original-source.tex, source lines 3941–3941. Navigation pointer invoked at proof line 3941; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $\mathcal{C}$ be a site. Let β be an ordinal. Let $\beta \to Sh(\mathcal{C})$, $\alpha \mapsto \mathcal{F}_\alpha$ be a system of sheaves over β. For $U \in \mathrm{Ob}(\mathcal{C})$ consider the canonical map*

$$\mathrm{colim}_{\alpha<\beta} \mathcal{F}_\alpha(U) \longrightarrow (\mathrm{colim}_{\alpha<\beta} \mathcal{F}_\alpha)(U)$$

If the cofinality of β is large enough, then this map is bijective for all U.

Proof. The left hand side is the value on U of the colimit $\mathcal{F}_{\mathrm{colim}}$ taken in the category of presheaves, see Section 00VB. Recall that $\mathrm{colim}_{\alpha<\beta} \mathcal{F}_\alpha$ is the sheafification $\mathcal{F}^\#_{\mathrm{colim}}$ of $\mathcal{F}_{\mathrm{colim}}$, see Lemma 00WI. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be an element of the set $\mathrm{Cov}(\mathcal{C})$ of coverings of $\mathcal{C}$. If the cofinality of β is larger than the cardinality of I, then we claim

$$H^0(\mathcal{U}, \mathcal{F}_{\mathrm{colim}}) = \mathrm{colim}\, H^0(\mathcal{U}, \mathcal{F}_\alpha) = \mathrm{colim}\, \mathcal{F}_\alpha(U) = \mathcal{F}_{\mathrm{colim}}(U)$$

The second and third equality signs are clear. For the first, say $s = (s_i) \in H^0(\mathcal{U}, \mathcal{F}_{\mathrm{colim}})$. Then for each i the element s_i comes from an element $s_{i,\alpha_i} \in \mathcal{F}_{\alpha_i}(U_i)$ for some $\alpha_i < \beta$. By the assumption on cofinality, we can choose $\alpha_i = \alpha$ independent of i. Then s_i and s_j map to the same element of $\mathcal{F}_{\alpha_{i,j}}(U_i \times_U U_j)$ for some $\alpha_{i,j} < \beta$. Since the cardinality if $I \times I$ is also less than the cofinality of β, we see that we may after increasing α assume $\alpha_{i,j} = \alpha$ for all i, j. This proves that the natural map $\mathrm{colim}\, H^0(\mathcal{U}, \mathcal{F}_\alpha) \to H^0(\mathcal{U}, \mathcal{F}_{\mathrm{colim}})$ is surjective. A very similar argument shows that it is injective. In particular, we see that $\mathcal{F}_{\mathrm{colim}}$ satisfies the sheaf condition for $\mathcal{U}$. Thus if the cofinality of β is larger than the supremum of the cardinalities of the set of index sets I of coverings, then we conclude. □